



DivQuant: Estimation of Species Richness and Entropy from Small Samples

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Motivation

Diversity measures

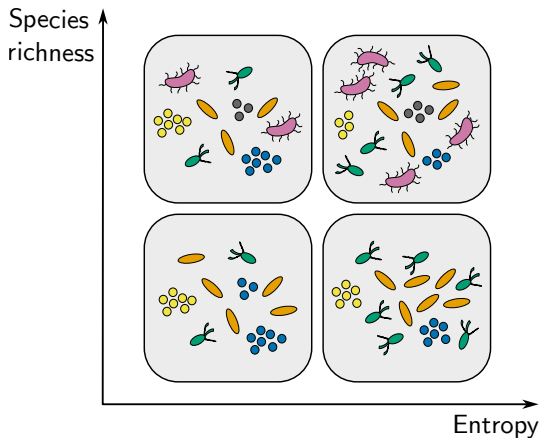
- Species richness: number of distinct elements
- Entropy (Shannon diversity): evenness
- Commonly used in microbiome analysis, ecology, immunology, etc.

Problem

- Sample richness underestimates the true population richness
- Richness increases with sampling depth
- How to compare samples?

Solution

- Rarefaction or **extrapolation**



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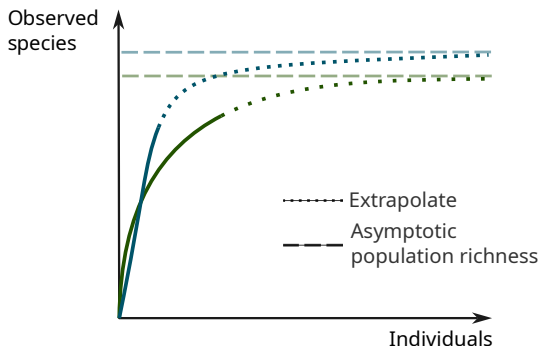
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Formal Problem Definition

Notation

- Species richness S
- Observed sample richness S_{obs}
- Sample size n for a population with N individuals
- Fingerprint $(f_k)_{k \geq 0}$
number of species seen k times
in the sample

It follows that

- $n = \sum_{k \geq 1} k f_k$
- $S_{obs} = \sum_{k \geq 1} f_k$

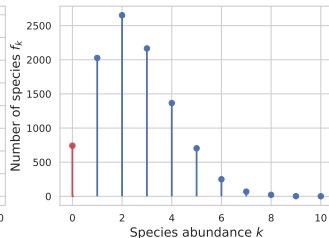
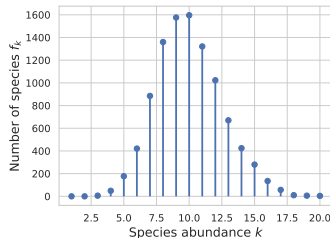
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Species richness estimation

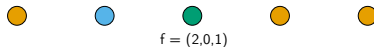
Species richness can be expressed by

$$S = \sum_{k \geq 0} f_k = S_{obs} + f_0.$$

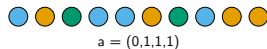
Therefore, estimating S is equivalent to estimating the unobserved f_0 from the observed f_1, f_2 , etc.

Sampling process

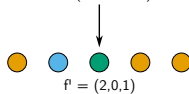
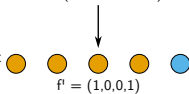
Observed fingerprint
($n = 5$)



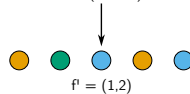
Population
($N = 10$)



Expected fingerprint
($n = 5$)



...



Idea: Estimate fingerprint of the total population such that the observed fingerprint is close to the expected sample fingerprint $f \approx f'$

Sampling process



$n = 10$ x_0 x_1 x_2 x_3 x_4 x_5 x_6 x_7 x_8 x_9 x_{10}

Assumptions

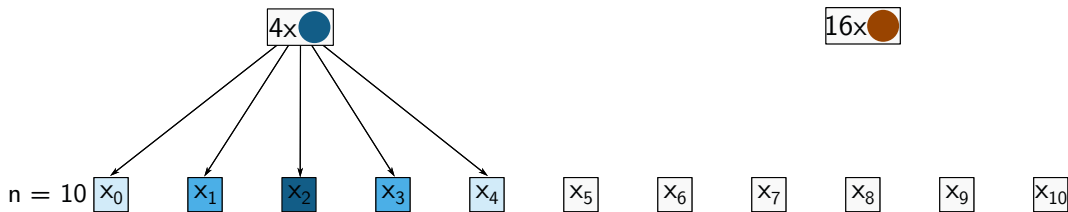
- Each individual is assigned to one class/species.
- Each sample reduces the population (**no** replacement).
- The sample is **not ordered**.

Hypergeometric distribution

For an urn with N balls, of which c are red, the probability to observe k red balls in a sample of n balls is

$$P_{c,k}^{N,n} = \frac{\binom{c}{k} \binom{N-c}{n-k}}{\binom{N}{n}}$$

Sampling process



Assumptions

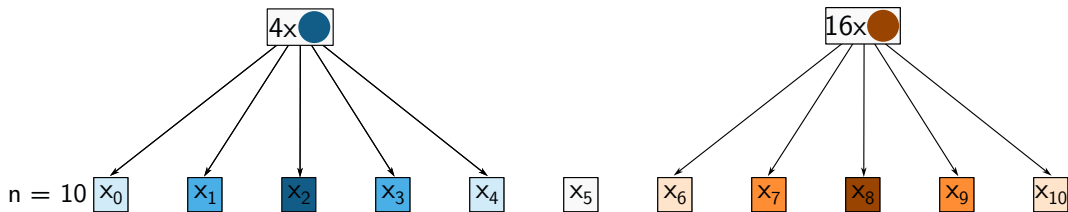
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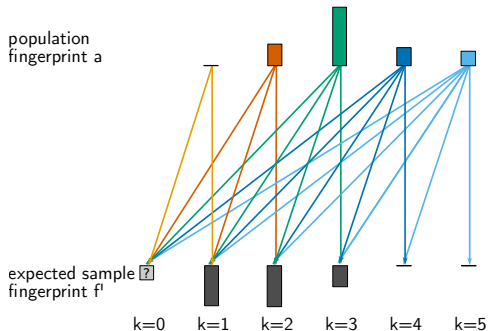
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Expected fingerprint

Expected fingerprint $f' = (f'_1, f'_2, \dots)$ for a population with fingerprint $a = (a_1, a_2, \dots)$

$$f'_k = \sum_{c \geq 1} a_c \cdot P_{c,k}^{N,n}$$



Quadratic optimization of DivQuant

$$\min \sum_{k=1}^K (f'_k - f_k)^2 / \max(f_k, 1)$$

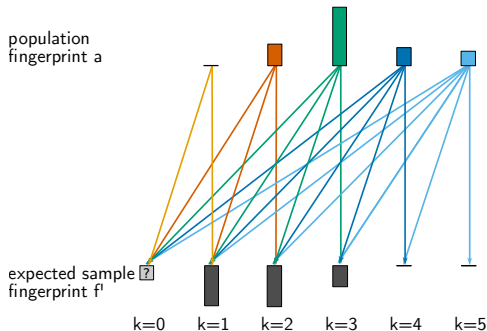
such that

$$f'_k = \sum_{c \geq 1} a_c \cdot P_{c,k}^{N,n} \quad (k = 0, \dots, K)$$

$$f'_0 = \sum_{c \geq 1} a_c - \sum_{k \geq 1} f_k$$

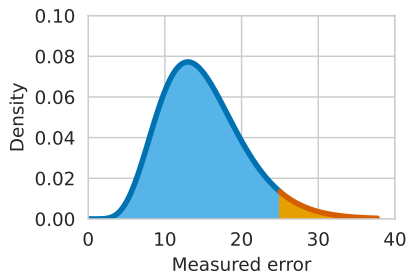
$$N = \sum_{c \geq 1} c \cdot a_c$$

$$a_c \geq 0 \quad (c = 1, \dots, C)$$



What is a plausible deviation between f and f' ?

- Neyman's χ^2 statistic^a: $\sum_{k=1}^K (f'_k - f_k)^2 / \max(f_k, 1)$
- Neyman's χ^2 statistic is asymptotically χ^2 distributed if it is approximately a sum of **squared independent standard normal variables**
- $f'_k = \sum_{c=k}^C a_c P_{c,k}^{N,n}$ can be approximated by a **Poisson** distribution
- Under Poisson assumption, the variance is $f_k \rightarrow$ standard normal approximation possible
- plausible deviation ε : quantile function of a χ^2 distribution with $K - 1$ degrees of freedom for the desired confidence level



^aJerzy Neyman (1949). "Contribution to the theory of the χ^2 test". In: **Proceedings of the Berkeley Symposium on Mathematical Statistics and Probability**, pp. 239–273.

Confidence interval

$$\min / \max \sum_{c=1}^C a_c$$

such that

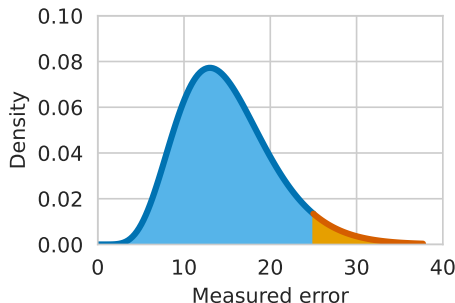
$$\varepsilon \geq \sum_{k=1}^K (f'_k - f_k)^2 / \max(f_k, 1)$$

$$f'_k = \sum_{c=k}^C a_c \cdot P_{c,k}^{N,n} \quad (k = 0, \dots, K)$$

$$f'_0 = \sum_{c=1}^C a_c - \sum_{k=1}^K f_k$$

$$N = \sum_{c=1}^C c \cdot a_c$$

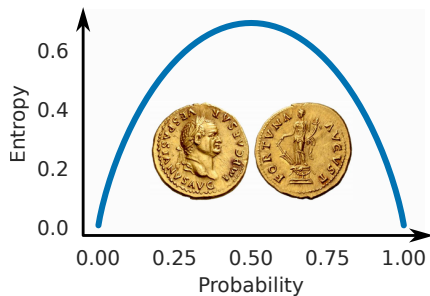
$$a_c \geq 0 \quad (c = 1, \dots, C)$$



Entropy estimation

Shannon diversity

- $H(X) = -\sum_{x \in X} p_x \log p_x$
- Quantifies randomness
- High values indicate large uncertainty
- Low values indicate small uncertainty



Assuming that $(a_c)_{c \geq 1}$ approximates the true population histogram

$$\hat{H}_a = -\sum_{c=1}^C a_c \frac{c}{N} \log \frac{c}{N}$$

is an accurate entropy estimate.
Minimizing or maximizing $\hat{H}_a$ yields confidence intervals.

Results

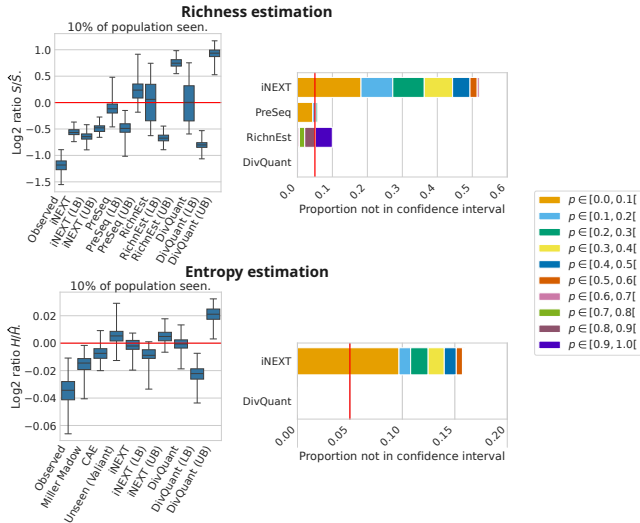
Richness estimation^a

- Point estimates of RichnEst and DivQuant are similar, but DivQuant's confidence intervals are wider.
- Confidence intervals of iNext and PreSeq do not cover the true value on many instances

Entropy estimation

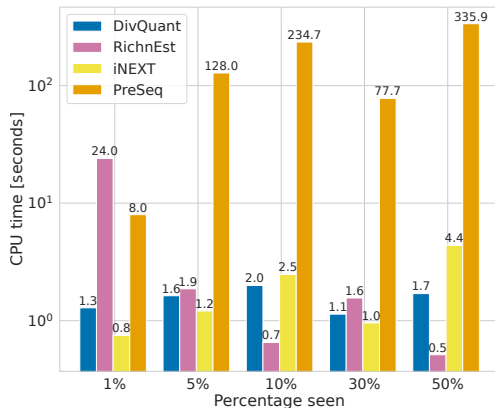
- DivQuant's confidence intervals contain the true entropy in more cases

^aJohanna E. Schmitz and Sven Rahmann (2025). "A comprehensive review and evaluation of species richness estimation". In: **Briefings in Bioinformatics** 26.2.

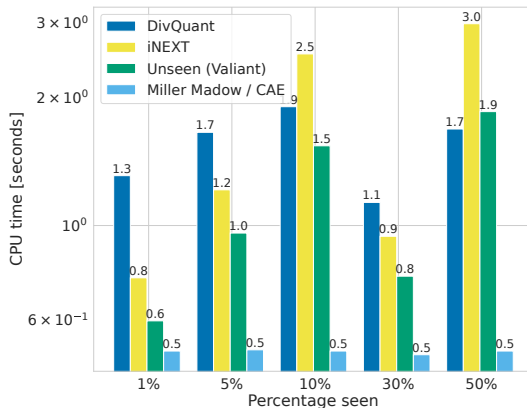


Running time comparison

Richness estimation



Entropy estimation



Tara Oceans sample TARA_000_DCM_0.22-1.6 ($N = 143\,169$)

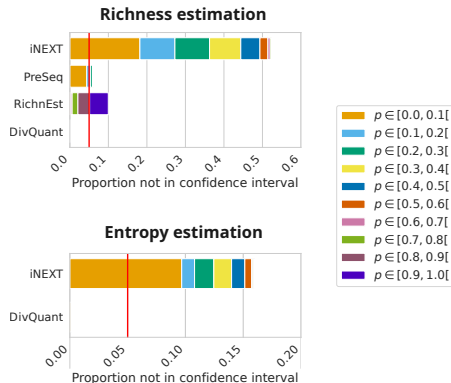
Summary

DivQuant

- is an **optimization-based** method to **extrapolate** species **richness** and **entropy**.
- yields **point estimates** and **confidence intervals**.
- attains close-to-nominal confidence intervals.

Future Work

- Tighten confidence intervals
- Perform more rigorous statistical analysis of approximations for χ^2 confidence intervals



References I

- Neyman, Jerzy (1949). “Contribution to the theory of the χ^2 test”. In: **Proceedings of the Berkeley Symposium on Mathematical Statistics and Probability**, pp. 239–273.
- Schmitz, Johanna E. and Sven Rahmann (2025). “A comprehensive review and evaluation of species richness estimation”. In: **Briefings in Bioinformatics** 26.2.
- Schröder, Christopher and Sven Rahmann (2015). “Efficient duplicate rate estimation from subsamples of sequencing libraries”. In: **PeerJ**.

Linear optimization of RichnEst¹

$$\min \sum_{k=1}^K \delta_k / \sqrt{f_k + 1}$$

such that

$$\delta_k \geq f'_k - f_k \quad (k = 1, \dots, K)$$

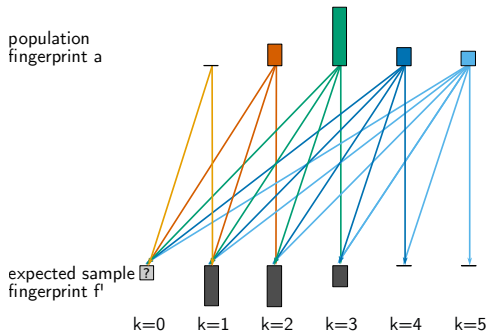
$$\delta_k \geq -f'_k + f_k \quad (k = 1, \dots, K)$$

$$f'_k = \sum_{c \geq 1} a_c \cdot P_{c,k}^{N,n} \quad (k = 0, \dots, K)$$

$$f'_0 = \sum_{c \geq 1} a_c - \sum_{k \geq 1} f_k$$

$$N = \sum_{c \geq 1} c \cdot a_c$$

$$a_c \geq 0 \quad (c = 1, \dots, C)$$



$$\text{Estimated richness } \hat{S} = \sum_{c \geq 1} a_c$$

¹Christopher Schröder and Sven Rahmann (2015). "Efficient duplicate rate estimation from subsamples of sequencing libraries". In: [PeerJ](#).

Appendix: Choose C and K

Let K_{obs} denote the largest count observed in the sample fingerprint f . Assuming that we have a species with abundance c in the population, the probability to observe a count larger than K_{obs} in our sample is given by $\sum_{k>K_{\text{obs}}} P_{c,k}^{N,n}$. Thus, the tail probability $\sum_{k>K_{\text{obs}}} P_{c,k}^{N,n} = 1 - \sum_{k=0}^{K_{\text{obs}}} P_{c,k}^{N,n}$ should be small. Given a threshold $t \in (0, 1)$, we therefore set

$$C := \min\{c \geq 1 : \sum_{k=0}^{K_{\text{obs}}} P_{c,k}^{N,n} \leq t\}.$$

Since the optimization program only measures the deviation between the observed and expected fingerprint up to K , we must ensure that essentially all the probability mass lies in the considered range. For a given C and a threshold $\tau \in (0, 1)$, we set

$$K := \min\{\kappa \geq 1 : \sum_{k=0}^{\kappa} P_{C,k}^{N,n} \geq \tau\}.$$

The observed fingerprint is padded with $K - K_{\text{obs}}$ trailing zeros.

Reducing the problem size

An abundant species' sample frequency is tightly concentrated around its expectation, so in the fingerprint f the values f_i for i near that frequency are small. We therefore say that a frequency k is **abundant** if its neighbourhood f is sparse:

$$\sum_{i \in [k - \lceil \sqrt{k} \rceil, k + \lceil \sqrt{k} \rceil]} f_i \leq 2\sqrt{k}.$$

We remove each **abundant** f_k from the fingerprint, give it an estimated population abundance of $k \cdot N/n$, and reduce n and N by $k \cdot f_k$ and $k \cdot f_k \cdot N/n$, respectively. After solving the reduced program, the empirical entries are added back to recover the full population fingerprint a .